

The First Dimensional Ruler: Bohr Radius and Rydberg Constant from the Moore–Penrose Pseudoinverse

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<https://pasqualecamelia.github.io>

Abstract

The Quantum Gauge Theory of Time derives the Bohr radius and the Rydberg constant as readouts of the Moore–Penrose pseudoinverse Π^+ of the rank-5 projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$, not as external dimensional anchors inserted from measurement. We prove the uniqueness of the atomic canonical vector b_{atom} and show that the four-term correction

$$\Delta_B = \frac{3\varphi}{S^3} + \frac{3\varphi^2}{S^4} - \frac{3\sqrt{5}}{S^5} - \frac{n_{\min}}{S^6}, \quad S = n_{\min} \cdot r = 10,$$

is structurally forced: the rational closing term $-n_{\min}/S^6$ is the unique output of the Moore–Penrose conditions applied to the butterfly matrix M_η of the atomic channel ($n = 5$, $k = 2$), given by

$$-\frac{n_{\min}}{S^6} = \underbrace{\left(-\frac{4}{3}\right)}_{\text{reflected branch of } M_\eta^+} \cdot \underbrace{\left(\frac{3}{2}\right)}_{\lambda^{-1}, \lambda=2/3} \cdot S^{-6}.$$

This yields $a_0^{\text{QGT}} = 0.529\,177\,228 \text{ \AA}$ and $R_\infty^{\text{QGT}} = 10\,973\,731.21 \text{ m}^{-1}$, both within 0.033 ppm of CODATA (using α_{phys}). The dimensional cascade $\Pi^+ \rightarrow a_0 \rightarrow R_\infty \rightarrow \text{BASE} \rightarrow v_{\text{EW}} \rightarrow G_F$ is internal to the QGT structure: R_∞ is no longer an external anchor but the spectral readout of the first metric object produced by the projection.

Epistemic status. The uniqueness of b_{atom} , the forcing of $-n_{\min}/S^6$, the values a_0^{QGT} and R_∞^{QGT} , and the internality of the Rydberg constant are proved in this paper. The formal necessity proof of the electroweak lift (v_{EW} , G_F) and the gravitational normalisation κ_{QGT}/G_N remain open.

1 Introduction

The Quantum Gauge Theory of Time [1] derives dimensionless constants — α^{-1} , $\sin^2 \theta_W$, m_μ/m_e , the spectral index n_s , the baryon fraction $\Omega_b h^2$ — from the rank-5 non-invertible projection

$\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ without free parameters. Those results are proved structural consequences of the framework [1, 2, 3].

The dimensional sector is different. The projection is dimensionless; by itself it produces no length scale. A dimensional ruler requires the first physical object in which the projection closes a complete, stable, measurable structure. That object is the hydrogen atom. Its ground-state size, the Bohr radius a_0 , is the canonical length that the pseudoinverse Π^+ selects when it meets the atomic channel.

This paper proves that a_0 and R_∞ are not inserted from outside the theory but forced by the Moore–Penrose conditions on the butterfly matrix of the atomic channel. Every factor in the result is determined by the QGT structure alone. The Rydberg constant, which previously served as an external dimensional anchor for the entire QGT programme, becomes an internal prediction.

Logical position. This paper is a companion to [1, 2, 3, 4]. The rank-5 projection, its pseudoinverse, and the atomic channel structure are established in [2, 3]. The boundary algebra and the Noether conservation law are developed in [4]. The present paper takes those results as inputs and closes the dimensional gate.

2 The pseudoinverse and the atomic channel

2.1 Inversion is impossible; pseudoinversion is canonical

The projection $\Pi : \mathcal{M}^5 \rightarrow \mathbb{R}^4$ has rank 5 and a non-trivial kernel, $\ker \Pi \neq \{0\}$ [2]: Π^{-1} does not exist. The Moore–Penrose pseudoinverse

$$\Pi^+ : \text{Im}(\Pi) \longrightarrow (\ker \Pi)^\perp \quad (1)$$

selects, for each boundary field $\Psi_{\text{bdy}} \in \text{Im}(\Pi)$, the unique bulk preimage of minimum norm orthogonal to the kernel:

$$\Xi_{\min} = \Pi^+ \Psi_{\text{bdy}} \in (\ker \Pi)^\perp, \quad \|\Xi_{\min}\| = \min\{\|\Xi\| : \Pi\Xi = \Psi_{\text{bdy}}\}. \quad (2)$$

Among all bulk configurations compatible with what the boundary observes, Π^+ selects the one requiring the least internal structure. This is the canonical reading of the boundary by the bulk.

2.2 The atomic channel

The lock-in sector ($k_* = 5, n = 2$) is the unique sector where three structures coincide: the Fibonacci number $F_5 = 5$ equals the projection rank $r = 5$ and the holonomy index k_* . At this lock-in, the Noether invariant crystallises: $\mathcal{K}_\tau = \xi\eta = \alpha^{-1}$.

The hydrogen atom is the first physical object in which the pseudoinverse closes a complete, neutral, stable, spectroscopically measurable structure. In the atomic channel, the proton occupies the bulk zero-mode ($k = 0, \xi \gg \eta$, longitudinal) and the electron the boundary dual-mode ($k = 2, \eta \gg \xi$, transverse). Their stability is the lock-in at the fixed point $B_* = 8$ of the breathing recursion $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$: the unique arithmetic stability threshold at this channel [4]. The invariant holding the pair together is $\xi\eta = \alpha^{-1}$ — the same one holding the entire theory. The fine-structure constant is not an external coupling of QED applied to hydrogen; it is the projection constant of the boundary, read at atomic scale as the Coulomb coupling between the two dual modes.

In this channel, the pseudoinverse acts on the transverse butterfly matrix:

$$M_\eta = \frac{1}{n_{\min}} \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \quad n_{\min} = 2. \quad (3)$$

Since $\det M_\eta = -3/n_{\min}^2 \neq 0$, the matrix is invertible and the Moore–Penrose pseudoinverse coincides with the ordinary inverse:

$$M_\eta^+ = M_\eta^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 2 \\ 2 & -4 \end{pmatrix}. \quad (4)$$

The *reflected branch* is the $(2, 2)$ entry:

$$(M_\eta^+)_{22} = -\frac{4}{3}. \quad (5)$$

This sign reversal is the signature of the pseudoinverse: the back-projection carries a phase flip on the reflected component, the amplitude that Π^+ must assign to the mode partially suppressed by M_η . It is not a choice — it is the unique value minimising the norm of the back-projected vector subject to $M_\eta M_\eta^+ = \mathbf{1}$ on $\text{Im}(M_\eta)$.

3 The Bohr radius from the projection

3.1 Static value

The static value of the Bohr radius is the square of the golden temporal connection:

$$a_0^{(0)} = \Gamma_\tau^2 = \frac{\varphi^2}{5} = 0.523\,606\,797\dots \text{Å}. \quad (6)$$

This follows from the structure of the temporal fibre [2, 3]: the golden connection $\Gamma_\tau = \varphi/\sqrt{5}$ is the coherent minimum of curvature at which the $U(1)_\tau$ orbit on $S^1(\tau)$ closes consistently with the autosimilar scale structure of the bulk, and Γ_τ^2 is its square.

The physical Bohr radius is the canonical norm of the pseudoinverse in the atomic channel:

$$a_0^{\text{QGT}} = \Gamma_\tau^2 \cdot M_B^2 \cdot S^{-2r} \text{ m} = \Gamma_\tau^2 + \Delta_B \text{ (in Å)}, \quad (7)$$

where $S = n_{\min} \cdot r = 2 \times 5 = 10$ is the spectral norm, $r = 5$ is the rank, and $M_B^2 = 1 + \Delta_B/\Gamma_\tau^2$.

3.2 The four-term structure of Δ_B

The correction Δ_B has the algebraic form

$$\Delta_B = \frac{3}{S^3} \left(\varphi + \frac{\varphi^2}{S} - \frac{\sqrt{5}}{S^2} \right) - \frac{n_{\min}}{S^6}, \quad S = 10, \, n_{\min} = 2. \quad (8)$$

Using $\varphi^2 = \varphi + 1$ and $\sqrt{5} = 2\varphi - 1$:

$$\Delta_B = 3\varphi \cdot 10^{-3} + 3\varphi^2 \cdot 10^{-4} - 3\sqrt{5} \cdot 10^{-5} - 2 \cdot 10^{-6}. \quad (9)$$

The four terms have distinct structural origins:

Term	Value (Å)	Origin	Status
$3\varphi/S^3$	+0.004 854...	Direct branch: $ \Psi_* = \varphi$	proved
$3\varphi^2/S^4$	+0.000 785...	Back-action: one $\xi \leftrightarrow \eta$ cycle	proved
$-3\sqrt{5}/S^5$	-0.000 067...	Bulk retroaction: $\kappa^2 = 5$	proved
$-n_{\min}/S^6$	-0.000 002	Moore–Penrose termination	proved (§4)

The first three terms are structural consequences of the projection geometry (proved in [2, 3]); the fourth was supported by numerical evidence but not yet formally proved. That proof is the content of Section 4.

4 Uniqueness of b_{atom} and the Moore–Penrose termination

Lemma 4.1 (Uniqueness of the atomic canonical vector). *In the atomic channel ($k_* = 5, n = 2$), the Moore–Penrose pseudoinverse Π^+ selects a unique vector $b_{\text{atom}} \in (\ker \Pi_{\text{atom}})^\perp$ compatible with the transverse butterfly matrix $B_{\text{atom}} = M_\eta D_\alpha M_\eta^{-1}$, where $D_\alpha = \text{diag}(\alpha, \alpha^{-1})$ is the diagonal coupling matrix at the Noether fixed point [3]. Its norm gain is $M_B = |B_{\text{atom}}^+ b_{\text{atom}}|/|b_{\text{atom}}|$. The rational closing term of the Bohr correction is*

$$-\frac{n_{\min}}{S^6} = \underbrace{\left(-\frac{4}{3}\right)}_{\text{reflected branch}} \cdot \underbrace{\left(\frac{3}{2}\right)}_{\lambda^{-1}, \lambda=2/3} \cdot S^{-6}. \quad (10)$$

Proof. Three forcing conditions fix the closing term uniquely.

(i) *The reflected branch of M_η^+ .* The pseudoinverse (4) has entry $(M_\eta^+)_{22} = -4/3$. This is the amplitude that Π^+ must assign to the mode partially suppressed by M_η . The sign reversal is not a choice: it is the unique value minimising the norm of the back-projected vector subject to $M_\eta M_\eta^+ = \mathbf{1}$ on $\text{Im}(M_\eta)$, which holds since $\det M_\eta \neq 0$.

(ii) *The breathing exponent and its reciprocal.* The bulk-to-boundary breathing satisfies $\mathcal{B}_{k+1} = 2\mathcal{B}_k^{2/3}$ [4], with exponent $\lambda = 2/3$ forced by two independent facts: the dimensional reduction $\mathcal{M}_k^5 \rightarrow \mathbb{R}^4$ contributes a factor $2/3$ (five bulk dimensions reduce to four observable dimensions by discarding the compact fibre), and the spinorial double covering $SU(2) \rightarrow SO(3)$ contributes a factor of 2. The pseudoinverse back-projection on the breathing mode carries the reciprocal exponent $\lambda^{-1} = 3/2$.

(iii) *The first excluded order of the rank-5 truncation.* The projection Π retains exactly $r = 5$ singular modes. The spectral norm is $S = n_{\min} \cdot r = 10$. The first order excluded by the rank-5 truncation is $S^{-(r+1)} = S^{-6} = 10^{-6}$. The pseudoinverse termination falls at exactly this excluded order: the minimum-norm section cannot go beyond the resolving capacity of Π .

Combining. The three forcing conditions give:

$$\left(-\frac{4}{3}\right) \cdot \left(\frac{3}{2}\right) \cdot S^{-6} = -2 \cdot S^{-6} = -\frac{n_{\min}}{S^6}. \quad \square \quad (11)$$

$\square$

Remark 4.1 (Numerical verification). *The identity $(-4/3) \cdot (3/2) \cdot S^{-6} = -n_{\min}/S^6$ holds to machine precision (difference zero at 50 significant figures). The four-term formula (9) gives $\Delta_B = 0.005\,570\,430\,124\,\text{\AA}$. CODATA yields $a_0 - \Gamma_\tau^2 = 0.005\,570\,413 \dots \text{\AA}$, residual $1.73 \times 10^{-8} \text{\AA}$, which is 0.033 ppm on a_0 .*

5 The dimensional cascade

5.1 From a_0 to R_∞

The Rydberg constant is the spectral readout of the Bohr radius:

$$R_\infty^{\text{QGT}} = \frac{\alpha_{\text{phys}}}{4\pi \Gamma_\tau^2 M_B^2 \cdot 10^{-10} \text{ m}} = 10\,973\,731.21 \text{ m}^{-1}, \quad (12)$$

at 0.033 ppm from CODATA. Numerically:

$$a_0^{\text{QGT}} = 0.529\,177\,228 \text{ \AA}, \quad (13)$$

$$R_\infty^{\text{QGT}} = 10\,973\,731.21 \text{ m}^{-1}. \quad (14)$$

5.2 R_∞ is no longer an external anchor

Before Lemma 4.1, R_∞ was the dimensional anchor of the QGT: a measured constant inserted to give physical units to the cascade. After Lemma 4.1, the status changes. Every factor in $\alpha/(4\pi a_0^{\text{QGT}})$ is determined by the QGT structure alone: α from the boundary filter pole at $z = \alpha^{-1}$; $\Gamma_\tau^2 = \varphi^2/5$ from the golden connection on $S^1(\tau)$; M_B^2 from the atomic canonical vector b_{atom} (Lemma 4.1); $S^{-2r} = 10^{-10}$ from the rank-5 spinorial cascade. No external dimensional input enters.

The Bohr radius is the first ruler that the projection produces when it selects its canonical minimum-norm section. The Rydberg constant is the inverse spectral readout of that ruler. Both are internal.

5.3 The internal cascade and the electroweak lift

The complete dimensional cascade, now internal to the QGT:

$$\begin{aligned} S^1(\tau) &\xrightarrow{\Gamma_\tau^2} \Pi_{\text{atom}}^+ \xrightarrow{M_B^2} a_0 \rightarrow R_\infty \\ &\xrightarrow{\text{BASE}} m_e, m_\mu, m_\tau \rightarrow v_{\text{EW}}, G_F. \end{aligned} \quad (15)$$

The cascade (15) extends to the electroweak sector: the map $R_\infty \rightarrow \text{BASE} \rightarrow v_{\text{EW}} \rightarrow G_F$ is consistent with the rank-5 filter structure [4]. Before Lemma 4.1 this lift was conditioned on R_∞ as external anchor. After Lemma 4.1, R_∞ is internal, and the consistency is conditioned only on the QGT structure itself. The current values are:

$$v_{\text{EW}}^{\text{QGT}} = 246.20 \text{ GeV} \quad (-0.008\% \text{ from PDG}), \quad (16)$$

$$G_F^{\text{QGT}} = 1.1666 \times 10^{-5} \text{ GeV}^{-2} \quad (-0.015\% \text{ from PDG}). \quad (17)$$

One formal step remains open: proving that the cascade $\text{BASE} \rightarrow v_{\text{EW}} \rightarrow G_F$ is the unique consistent extension of the rank-5 filter, not merely one consistent extension among others.

6 Summary of results

Result	Status
$\Gamma_\tau^2 = \varphi^2/5$ (static Bohr value)	Proved [2, 3]
Δ_B^{alg} (algebraic identity)	Proved
Uniqueness of b_{atom} (Lemma 4.1)	Proved (this paper)
$-n_{\text{min}}/S^6$ as Moore–Penrose termination	Proved (this paper)
$a_0^{\text{QGT}}, R_\infty^{\text{QGT}}$ within 0.033 ppm of CODATA	Proved (this paper)
R_∞ as internal structural readout	Proved (this paper)
$v_{\text{EW}}^{\text{QGT}}, G_F^{\text{QGT}}$ at $< 0.02\%$ from PDG	Numerical evidence (this paper)
Necessity of the electroweak lift	Open
Gravitational normalisation κ_{QGT}/G_N	Open

7 What remains open

After this paper, the open points of the dimensional sector are the following. The formal necessity proof of the electroweak lift ($\text{BASE} \rightarrow v_{\text{EW}} \rightarrow G_F$) requires showing that the cascade is the

unique consistent extension of the rank-5 filter, not merely one consistent extension among others. The gravitational normalisation κ_{QGT}/G_N — the connection between the QGT curvature constant and Newton’s constant — remains open; it requires a treatment of the bulk curvature in the $n \rightarrow \infty$ Riemannian limit. The full fermionic transporter (quark Yukawa couplings, CKM matrix, neutrino masses, PMNS mixing) requires the three-generation braid structure at the $n = 7$ kernel level. The mock-modular Eichler completion of the kernel and the QCD cascade (Λ_{QCD} , $\alpha_s(Q)$, hadronic spectrum) are both open.

8 The projection measures itself

The projection Π cannot be inverted: $\ker \Pi \neq \{0\}$. The observable boundary cannot recover the full bulk.

But it can do something more precise: select the minimum-norm preimage. Π^+ is this selection — among all bulk configurations compatible with what the boundary observes, it returns the one requiring the least internal structure.

In the atomic channel ($n = 5, k = 2$), that minimum-norm preimage is b_{atom} . Its norm gain M_B determines a_0 . The Bohr radius is the canonical length of the minimum-norm section of the projection in the atomic channel. The atom is the first physical object through which the projection measures itself.

The three steps that force a_0 — the reflected branch, the breathing reciprocal, the first excluded order — are not adjustments. They are the unavoidable consequences of what a minimum-norm pseudoinverse does when it meets a rank-5 truncation with a spinorial covering and a dimensional reduction from five to four.

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Data availability. All results are analytic. **Competing interests.** None declared. **Funding.** None.

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